

Chapter 12: Project Groups - Literature Review

Finding and Evaluating Research

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1 Overview

This reading prepares you for the first project group session, where you'll begin your **literature review**. A literature review isn't just a summary of papers — it's a critical synthesis that establishes why your research question matters and what's already known about it.

i What You'll Learn

By the end of this lab, you'll be able to:

- Develop effective database search strategies
- Evaluate the quality and relevance of research papers
- Organise your sources systematically
- Begin drafting your introduction section

2 Why Literature Reviews Matter

A good literature review does several things:

1. **Establishes context:** Shows what's already known in the field
2. **Identifies gaps:** Reveals what hasn't been studied or remains unclear
3. **Justifies your study:** Explains why your research question is worth investigating
4. **Informs methodology:** Shows how others have approached similar questions

! Your Goal

Your introduction should create a “funnel” — starting broad (the general topic), narrowing to specific previous findings, and concluding with your specific research question and hypothesis.

3 Search Strategies

3.1 Choosing Keywords

Start by identifying the key concepts in your research question. For example:

Research question: “Does social media use affect sleep quality in university students?”

Key concepts:

- Social media use
- Sleep quality
- University students / young adults

3.2 Building Search Terms

For each concept, brainstorm:

- **Synonyms:** “social media” OR “social networking sites” OR “Facebook” OR “Instagram”
- **Broader terms:** “internet use” includes social media
- **Narrower terms:** “Instagram use” is more specific than “social media”

3.3 Using Boolean Operators

Operator	Effect	Example
AND	Narrows search (both terms required)	social media AND sleep
OR	Broadens search (either term)	Facebook OR Instagram
NOT	Excludes terms	social media NOT addiction
" "	Exact phrase	“sleep quality”
*	Wildcard	adolescen* finds adolescent, adolescents, adolescence

3.4 Example Search String

```
("social media" OR "social networking") AND  
("sleep quality" OR "sleep duration" OR insomnia) AND  
(student* OR "young adult*" OR undergraduate*)
```

4 Key Databases for Psychology

Database	Strengths	Access
PsycINFO	Psychology-specific, best for theoretical articles	Library
PubMed	Biomedical focus, good for clinical topics	Free
Web of Science	Cross-disciplinary, citation tracking	Library
Google Scholar	Broad coverage, easy to use	Free

Database	Strengths	Access
Scopus	Wide coverage, good metrics	Library

Start with PsycINFO

For psychology research, start with PsycINFO. It's specifically designed for psychological literature and uses standardised subject terms.

5 Evaluating Sources

Not all papers are equally useful. Ask yourself:

5.1 Is it relevant?

- Does it address your research question directly?
- Does it study a similar population?
- Are the methods applicable to your study?

5.2 Is it credible?

- Is it peer-reviewed? (Check: is it in a journal?)
- Who are the authors? (Do they have relevant expertise?)
- Is it published in a reputable journal?
- Has it been cited by others?

5.3 Is it current?

- How old is the research?
- Is this a rapidly changing field where older studies might be outdated?
- Are there more recent replications or meta-analyses?

Red Flags

Be cautious of:

- Non-peer-reviewed sources (blogs, news articles, opinion pieces)
- Predatory journals (poor quality control)
- Studies with very small samples
- Single studies that haven't been replicated
- Sources that cite only themselves

6 Reading Research Papers Efficiently

You don't need to read every paper in full. Use a strategic approach:

6.1 1. Screen by Abstract

Read the abstract first. Does this paper:

- Address your topic?
- Use relevant methods?
- Study a similar population?

If not, move on.

6.2 2. Skim the Introduction and Discussion

- **Introduction:** What's their research question? What do they cite?
- **Discussion:** What did they find? What do they recommend for future research?

6.3 3. Deep Read Key Papers

For highly relevant papers, read carefully:

- What exactly did they measure and how?
- What were the limitations?
- How does this inform your own study?

7 Organising Your Sources

7.1 Reference Management

Use a reference manager to:

- Store PDFs and notes
- Generate citations automatically
- Create bibliographies

Popular options:

Tool	Pros	Cons
Zotero	Free, browser integration, collaborative	Learning curve
Mendeley	Free, good PDF reader	Owned by publisher
EndNote	Powerful	Expensive

7.2 Taking Notes

For each paper, record:

- Full citation (author, year, title, journal)
- Key findings relevant to your question
- Methods used (sample size, measures, design)
- Strengths and limitations
- How it relates to your study

7.3 Creating a Literature Matrix

A simple table helps you synthesise:

Study	Sample	Method	Key Finding	Relevance to Your Study
Smith (2020)	200 students	Survey	SM use → poor sleep	Direct
Jones (2019)	150 adults	Experiment	Blue light → delayed sleep	Mechanism

8 From Notes to Narrative

Your literature review should tell a story, not just list studies. Compare:

Poor (list-style): > Smith (2020) found that social media use affected sleep. Jones (2019) also studied social media and sleep. Brown (2021) looked at screen time.

Better (narrative): > Recent research has established a link between social media use and sleep quality. Smith (2020) found that students who used social media for more than two hours daily reported significantly worse sleep quality. This relationship may be explained by blue light exposure (Jones, 2019) or by the emotionally arousing nature of social content (Brown, 2021).

8.1 Synthesis Tips

- **Group by theme**, not by paper
- **Compare and contrast** findings across studies
- **Identify patterns** and inconsistencies
- **Build toward** your specific question

9 Common Errors and How to Avoid Them

9.1 Relying too heavily on Google Scholar

Problem: Google Scholar includes non-peer-reviewed sources and doesn't filter by quality.

Solution: Use Google Scholar to find papers, but verify they're peer-reviewed. Cross-reference with PsycINFO.

9.2 Using only recent papers

Problem: Missing foundational/classic studies that established the field.

Solution: Follow citation trails — what papers do recent articles cite? Include seminal works.

9.3 Over-quoting

Problem: Too many direct quotes make your writing feel disjointed.

Solution: Paraphrase whenever possible. Direct quotes should be rare and used for particularly important or well-phrased points.

9.4 Not being critical

Problem: Treating all studies as equally valid.

Solution: Note limitations. A study with $n = 20$ is less reliable than $n = 200$. Self-report measures have different limitations than objective measures.

10 Preparation Checklist

Before arriving at the lab:

Before You Arrive

- ☐ Have your research question clearly defined
- ☐ Create a list of key search terms and synonyms
- ☐ Access the library databases (test login beforehand)
- ☐ Install or access a reference manager (Zotero recommended)
- ☐ Review your hypothesis and what evidence would support/refute it

10.1 Questions to Discuss with Your Group

- What are the key variables in our research question?
- What search terms will capture relevant studies?
- How will we divide the search work?
- How will we share and organise our findings?

10.2 What to Bring

- Laptop with library database access
- Reference manager installed/ready
- Your research question and hypothesis
- Notes from previous lab sessions

11 Summary

Task	Key Points
Searching	Use Boolean operators, try multiple databases
Evaluating	Check peer-review status, author credibility, recency
Reading	Screen by abstract, skim for relevance, deep-read key papers
Organising	Use reference manager, create literature matrix
Writing	Synthesise by theme, build a narrative, be critical

12 Additional Resources

- Goldsmiths Library Guide to PsycINFO — Database access and tutorials
- Purdue OWL: Literature Reviews — Writing guidance
- Zotero Quick Start Guide — Reference management setup
- Boolean Operators Explained — Search strategy help